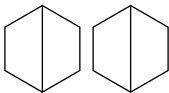


BIOZOLE
Fluconazole



COMPOSITION
Each capsule contains : Fluconazole 50 mg, 100 mg, 150 mg, 200 mg
DOSAGE FORM AVAILABLE
CAPSULE
PRODUCT DESCRIPTION
Biozole capsule 50 mg :
White powder filled in light blue (cap)/ white (body), opaque, hard gelatin No.4 capsules printing in black Biolab's logo on one side and "BIOLAB" on the other.
Biozole capsule 100 mg :
White powder filled in blue (cap)/ white (body), opaque, hard gelatin No.2 capsules printing in black Biolab's logo on one side and "BIOLAB" on the other.
Biozole capsule 150 mg :
White powder filled in pink (cap)/ pink (body), opaque, hard gelatin No.1 capsules printing in black "BIOZOLE" on one side and "150" on the other.
Biozole capsule 200 mg :
White powder filled in purple (cap)/ white (body), opaque, hard gelatin No.0 capsules printing in black Biolab's logo on one side and "BIOLAB" on the other.

ACTIONS
Fluconazole, a member of a new class of triazole antifungal agents, is a potent and specific inhibitor of fungal sterol synthesis.
Pharmacokinetics
Absorption : The pharmacokinetics of fluconazole are similar following IV or oral administration. The drug is rapidly and almost completely absorbed from the GI tract, and there is no evidence of first-pass metabolism. Oral bioavailability of fluconazole exceeds 90% in healthy, fasting adults ; peak plasma concentrations of the drug generally are attained within 1-2 hours after oral administration.
Distribution : Fluconazole is widely distributed into body tissues and fluids following oral or IV administration. Fluconazole, unlike some azole-derivative antifungal agents (e.g., itraconazole, ketoconazole), distributes readily into CSF following oral or IV administration ; CSF concentrations of fluconazole may be 50-94% of concurrent plasma concentrations regardless of the degree of meningeal inflammation.
Elimination : The plasma elimination half-life of fluconazole in adults with normal renal function is approximately 30 hours. In patients with impaired renal function, plasma concentrations of fluconazole are higher and the half-life prolonged ; elimination half-life of the drug is inversely proportional to the patients's creatinine clearance. Fluconazole is removed by hemodialysis and peritoneal dialysis.

TOXICOLOGY
There was no evidence of mutagenicity when fluconazole was tested with or without metabolic inactivation in 4 strains of Salmonella typhimurium or in the mouse lymphoma L5178Y system. In addition, there was no evidence of chromosomal mutations in vivo on murine bone marrow cells following administration of fluconazole or in vitro on human lymphocytes exposed to fluconazole concentrations of 1 mg/ml. There was no evidence of carcinogenicity in studies in mice and rats receiving oral fluconazole dosages of 2.5-10 mg/kg daily (approximately 2-7 times the usual human dosage) for 24 months. However, there was an increased incidence of hepatocellular adenomas in male rats receiving an oral fluconazole dosage of 5 or 10 mg/kg daily.
INDICATIONS
Fluconazole is indicated for the treatment of the following conditions :
Cryptococcosis, including cryptococcal meningitis and infections of other sites (e.g. pulmonary, cutaneous). Normal hosts, and patients with AIDS, organ transplants or other causes of immunosuppression may be treated. Fluconazole can be used as maintenance therapy to prevent relapse of cryptococcal disease in patients with AIDS.

Systemic candidiasis including candidemia, disseminated candidiasis and other forms of invasive candidal infection of the peritoneum, endocardium and pulmonary and urinary tracts. Patients with malignancy, in intensive care units, receiving cytotoxic or immunosuppressive therapy, or with other factors predisposing to candidal infection may be treated.
Mucosal candidiasis those including oropharyngeal, non-invasive bronchopulmonary infections, candiduria, mucocutaneous and chronic oral atrophic candidiasis (denture sore mouth). Normal hosts and patients with compromised immune function may be treated.
Vaginal candidiasis, acute or recurrent.
Prevention of fungal infection in patients with malignancy who are predisposed to such infections as a result of cytotoxic chemotherapy or radiotherapy.
Dermatomycosis including tinea pedis, tinea corporis, tinea cruris, tinea versicolor and candida infections.

DOSAGE AND ADMINISTRATIONS
Route of administration : Oral
The daily dose of fluconazole should be based on the nature and severity of the fungal infection. Most cases of vaginal candidiasis respond to single-dose therapy. Therapy for those types of infections requiring multiple-dose treatment should be continued until clinical parameters or laboratory tests indicate that active fungal infection has subsided. An inadequate period of treatment may lead to recurrence of active infection.
Patients with AIDS and cryptococcal meningitis or recurrent oropharyngeal candidiasis usually require maintenance therapy to prevent relapse.

Adults :
For cryptococcal meningitis and cryptococcal infections at other sites, the usual dose is 400 mg on the first day followed by 200-400 mg once daily. Duration of treatment for cryptococcal meningitis will depend on the clinical and mycological response, but is usually at least 6-8 weeks for cryptococcal meningitis.
For the prevention of relapse of cryptococcal meningitis in patients with AIDS, after the patient receives a full course of primary therapy fluconazole may be administered indefinitely at a daily dose of 200 mg
For candidemia, disseminated candidiasis and other invasive candidal infections, the usual dose is 400 mg on the first day followed by 200 mg daily. Depending on the clinical response, the dose may be increased to 400 mg daily. Duration of treatment is based upon the clinical response.
For oropharyngeal candidiasis, the usual dose is 50 mg once daily for 7-14 days. If necessary, treatment can be continued for longer periods in patients with severely compromised immune function.
For atrophic oral candidiasis associated with dentures, the usual dose is 50-100 mg once daily for 7-14 days administered concurrently with local antiseptic measures to the denture.
For other candidal infections of mucosa, (except vaginal candidiasis), e.g. oesophagitis, non-invasive bronchopulmonary infections, candiduria, mucocutaneous candidiasis, etc, the usual effective dose is 50 mg daily, given for 14-30 days.

In unusually difficult cases of mucosal candidal infections, the dose may be increased up to 100 mg daily.
For vaginal candidiasis, fluconazole 150 mg should be administered as a single oral dose.
For the prevention of candidiasis, recommended dosage is 50-400 mg once daily, based on the patient's risk for developing fungal infection. For patients at high risk of systemic infection, e.g. patients who are anticipated have profound neutropenia, the recommended daily dose is 400 mg once daily. Fluconazole administration should start several days before the anticipated onset of neutropenia and continuing for 7 days after the neutrophil count rises >1000 cells/mm³.
For dermal infections including tinea pedis, corporis, cruris and candida infections the recommended dosage is 150 mg once weekly or 50 mg once daily. Duration of treatment is normally 2-4 weeks but tinea pedis may require treatment for up to 6 weeks. For tinea versicolor, the recommended dose is 50 mg once daily for 2-4 weeks.

As with similar infections in adults, the duration of treatment is based on clinical and mycological response. Fluconazole is administered as a single dose each day.
In children : As with similar infections in adults, the duration of treatment is based on the clinical and mycological response. The maximum adult daily dosage should not be exceeded in children. Fluconazole is administered as a single dose each day.
For oropharyngeal candidiasis : Recommended dose is 3 mg/kg daily. A loading dose of 6 mg/kg may be used on the first day to achieve steady-state levels more rapidly.
For treatment of cryptococcal meningitis : Recommended dose is 6-12 mg/kg daily, depending on the severity of the disease.
Prevention of fungal infections in immunocompromised patients considered at risk as a consequence of neutropenia following cytotoxic chemotherapy or radio therapy : 3-12 mg/kg daily, depending on the extent and duration of the induced neutropenia (see adult dosing).

Children <4 weeks :
Neonates excrete fluconazole slowly. In the first weeks of life, the same mg/kg dosing as in order children should be used but administered every 72 hours. During weeks 2-4 of life, the same dose should be given every 48 hours.
Use in Elderly :
When there is no evidence of renal impairment, normal dosage recommendations should be adopted. For patients with renal impairment (creatinine clearance <50 ml/min) the dosage schedule should be adjusted as described below.
In patients with renal impairment : Fluconazole is predominantly excreted in the urine as unchanged drug. No adjustments in single-dose therapy are necessary. In patients with impaired renal function who will receive multiple doses of fluconazole, an initial loading dose of 50 mg to 400 mg should be given. After the loading dose, the daily dose (according to indication) should be based on the following table.

Table with 2 columns: Creatinine Clearance (ml/min), Percent of Recommended Dose. Rows show dosages for >50, 11-50, and Patients receiving regular dialysis.

OVERDOSAGE
In the event of overdosage, symptomatic treatment (with supportive measures and gastric lavage if necessary) may be adequate.
Fluconazole is largely excreted in the urine ; forced volume diuresis would probably increase the elimination rate. A 3-hr hemodialysis session decreases plasma levels be approximately 50%.
CONTRAINDICATIONS
Fluconazole should not be used in patients with known sensitivity to the drug or to related azole compounds.
PRECAUTIONS AND WARNINGS
Very rarely, patients who died with severe underlying disease and who had received multiple-dose fluconazole had post-mortem findings which included hepatic necrosis. These patients were receiving multiple concomitant medications, some known to be potentially hepatotoxic and/or had underlying diseases which could have caused the hepatic necrosis. Consequently, because a causal relationship with fluconazole can not be excluded, in those patients in whom a significant rise of liver enzymes occurs, the risk-benefit ratio of continued fluconazole treatment should be assessed. In rare cases, as with other azoles, anaphylaxis has been reported.

Use During Pregnancy
There have been reports of spontaneous abortion and congenital abnormalities in infants whose mothers were treated with 150 mg of fluconazole as a single or repeated dose in the first trimester. Use in pregnancy should be avoided expect in patients with severe or potentially life-threatening fungal infections in whom BIOZOLE may be used if the anticipated benefit outweighs the possible risk to the fetus. If this drug is used during pregnancy, or if the patient becomes pregnant while taking the drug, the patient should be informed of the potential hazard to the fetus.
Effective contraceptive measures should be considered in women of child-bearing potential and should continue throughout the treatment period and for approximately 1 week (5 to 6 half-lives) after the final dose.
There have been reports of multiple congenital abnormalities in infants whose mothers were treated with high-dose (400 mg/day to 800 mg/day) fluconazole therapy for coccidioidomycosis (an unapproved indication). The relationship between fluconazole use and these events is unclear. Adverse fetal effects have been seen in animals only at high-dose levels associated with maternal toxicity. There were no fetal effects at 5 mg/kg or 10 mg/kg; increases in fetal anatomical variants (supernumerary ribs, renal pelvis dilation) and delays in ossification were observed at 25 mg/kg and 50 mg/kg and higher doses. At doses ranging from 80 mg/kg (approximately 20-60 times the recommended human dose) to 320 mg/kg, embryolethality in rats were increased and fetal abnormalities included wavy ribs, cleft palate and abnormal craniofacial ossification. Case reports describe a distinctive and a rare pattern of birth defects among infants whose mothers received high dose (400-800 mg/day) fluconazole during most or all of the first trimester of pregnancy. The features seen in these infants include brachycephaly, abnormal facies, abnormal calvarial development, cleft palate, femoral bowing, thin ribs and long bones, arthrogryposis, and congenital heart disease.

Use During Lactation
Fluconazole is found in human breast milk at concentrations similar to plasma. Breast-feeding may be maintained after a single dose of 150 mg fluconazole. Breast-feeding is not recommended after repeated use or after high-dose fluconazole.
Driving/Use of Machinery : Experience with fluconazole indicates that therapy is unlikely to impair a patient's ability to drive or use machinery.I

SIDE EFFCETS
Fluconazole is generally well tolerated. The most common side effects associated with fluconazole are symptoms related to the gastrointestinal tract. These include nausea, headache, abdominal pain, diarrhea and flatulence.
After gastro-intestinal symptoms, the second most commonly observed side effect was rash. In some patients, particularly those with serious underlying diseases e.g. AIDS and cancer, changes in renal and hematological function test results and hepatic abnormalities (see warnings) have been observed during treatment with fluconazole and comparative agents, but the clinical significance and relationship to treatment is uncertain.
Patients with AIDS are more prone to the development of severe cutaneous reactions to many drugs. A small number of AIDS patients have developed such reaction, usually while receiving fluconazole concomitantly with other agents known to be associated with exfoliative cutaneous reactions. However, a definite causal relationship between exfoliative skin eruptions and the drug has not been established, since most patients were receiving multiple drugs concomitantly with fluconazole. If a rash develops in a patient treated for a superficial fungal infection which is considered attributable to fluconazole, further therapy with this agent should be discontinued. If patients with invasive/systemic fungal infections develop rashes, they should be monitored closely and fluconazole discontinued if bullous lesions or erythema multiforme develop. In rare cases, as with other azoles, anaphylaxis has been reported.

DRUG INTERACTIONS
Anticoagulants : Fluconazole increased the prothrombin time after warfarin administration in healthy males. Though the magnitude of change was small (12%), careful monitoring of prothrombin time in patients receiving coumarin-type anticoagulants is recommended.
Sulfonylureas : Fluconazole has been shown to prolong the serum half-life of concomitantly administered oral sulfonylureas (chlorpropamide, glibenclamide, glipizide and tolbutamide) in healthy volunteers. Fluconazole and oral sulfonylureas may be co-administered to diabetic patients, but the possibility of a hypoglycemic episode should be borne in mind.
Hydrochlorothiazide : In a kinetic interaction study, co-administration of multiple-dose hydrochlorothiazide to healthy volunteers receiving fluconazole increased plasma concentrations of fluconazole by 40%. An effect of this magnitude should not necessitate a change in the fluconazole dose regimen in subjects receiving concomitant diuretics, although the prescriber should bear it in mind.

Phenytoin : Concomitant administration of fluconazole and phenytoin may increase the levels of phenytoin to a clinically significant degree. If it is necessary to administer both drugs concomitantly, phenytoin levels should be monitored and phenytoin does adjusted to maintain therapeutic levels.
Oral contraceptives : Two kinetic studies with a combined oral contraceptive have been performed using multiple doses of fluconazole. There were no relevant effects on either hormone level in the 50 mg fluconazole study while at 200 mg daily, the AUCs of ethinyl estradiol and levonorgestrel were increased, 40% and 24%, respectively. Thus, multiple-dose use of fluconazole at these doses is unlikely to have an effect on the efficacy of the combined oral contraceptive.
Rifampicin : Concomitant administration of fluconazole and rifampicin resulted in a 25% decrease in the AUC and 20% shorter half-life of fluconazole. In patients receiving concomitant rifampicin, an increase of the fluconazole dose should be considered.
Zidovudine : Concomitant administration of fluconazole appears to interfere with the metabolism and clearance of zidovudine. In one study in men with HIV infections who received zidovudine (200 mg every 8 hours) alone or in conjunction with fluconazole (400 mg daily), the AUC of zidovudine was increased 74% (range : 20-173%), peak serum zidovudine concentrations were increased 84% (range : -1 to 227%) and the terminal elimination half-life of the drug was increased 128% (range : -4 to 189%) in patients receiving concomitant fluconazole. Although the clinical importance of this effect is unknown, it has been suggested that patients receiving concomitant zidovudine and fluconazole therapy by monitored closely for zidovudine-associated adverse effects.

Cyclosporin : A kinetic study in renal transplant patients found fluconazole 200 mg daily to slowly increase cyclosporin concentrations. However, in another multiple-dose study with 100 mg daily, fluconazole did not affect cyclosporin levels in patients with bone marrow transplants. Cyclosporin plasma concentration monitoring in patients receiving fluconazole is recommended.
Terfenadine : The possibility that fluconazole may interact with terfenadine (no longer commercially available in the US), resulting in potentially serious adverse cardiovascular effects, should be considered. Prolongation of the QT interval and QT interval corrected for rate (QTc) and rarely, serious cardiovascular effects, including arrhythmias (e.g., ventricular tachycardia, atypical ventricular tachycardia [torsades de pointes, ventricular fibrillation]), cardiac arrest, palpitations, hypotension, dizziness, syncope and death, have been reported in patients receiving recommended dosage of terfenadine.

Theophylline : In a placebo-controlled interaction study, the administration of fluconazole 200 mg for 14 days resulted in an 18% decrease in the mean plasma clearance rate of theophylline. Patients who are receiving high doses theophylline or who are otherwise at increased risk for theophylline toxicity should be observed for signs of theophylline toxicity while receiving fluconazole, and therapy modified appropriately if signs of toxicity develop.

STORAGE
Store in a dry place at temperature not exceeding 30°C.
Keep medicine out of reach of children.
Jauhkan daripada kanak-kanak.
The shelf-life period is 3 years.
SUPPLY
Biozole capsule 50 mg : Blister pack in printing box of 10x10's
Biozole capsule 100 mg : Blister pack in printing box of 5x10's
Biozole capsule 150 mg : Blister pack in printing box of 50x1's
Biozole capsule 200 mg : Blister pack in printing box of 5x10's

Date of revision : November 2017

Manufactured by
BIOLAB CO., LTD.
625 Bangpoo Industrial Estate,
Samutprakarn, Thailand.

EEC030/C

FRONT

Form containing product information, signatures, and specifications for EEC030/C. Includes fields for Code, Draft Revision, Final Artwork Revision, Name, Remark, Design by, Review by, Approved by, and Specification details.

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